



Ashmit Khandelwal

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Education

Carnegie Mellon University

M.S. MACHINE LEARNING

Pittsburgh, PA

Aug 2026 - Present

Birla Institute of Technology and Science, Pilani

B.E. COMPUTER SCIENCE, MINOR IN DATA SCIENCE

India

2020 - 24

GPA: 9.5/10.0

Work Experience

Research Fellow - Microsoft Research

Bangalore, India

ADVISORS: [NAGARAJAN NATARAJAN](#), [AMIT SHARMA](#)

Jul 2024 - June 2026

- Developed interwhen, a test-time verification framework to asynchronously monitor and steer LLM agent trajectories. Spans code generation, arithmetic, logical, spatial, and agentic settings.
- Formally defined Deep Research (DR), a previously under-defined class of LLMs for large-scale search and reasoning. Introduced LiveDR-Bench, one of the first benchmarks to evaluate DR, highlighting performance gaps in DR systems from OpenAI, Perplexity, and Gemini.

Research Intern - Adobe

Noida, India

ADVISORS: [YAMAN KUMAR \(ADOBE\)](#), [CHANGYOU CHEN \(SUNY BUFFALO\)](#)

Summer 2023

- Modeled human behavior from YouTube videos to predict rewatchability, engagement metrics, and audience sentiment at scale across diverse content domains.
- Designed and trained multimodal LLM architectures that integrated video and textual representations, achieving significant performance gains and surpassing GPT-4 on multiple behavior prediction tasks.
- Built pipelines for video, audio, and text scraping and processing, automating collection of terabytes of web data.

Publications

* = equal contribution

Large Content And Behavior Models To Understand, Simulate, And Optimize Content And Behavior

Spotlight at ICLR

A KHANDELWAL*, A AGRAWAL*, A BHATTACHARYYA*, YK SINGLA*, ET AL.

2024

Large Content and Behavior Models (LCBMs) reintroduce behavior tokens into LLM training data to simulate and explain human audience behavior in response to different forms of content. These models generalize across content types and adapt to diverse behavior domains, demonstrated on the newly introduced Content Behavior Corpus (CBC).

Characterizing Deep Research: A Benchmark and Formal Definition

ICLR

A JAVA*, A KHANDELWAL*, S MIDIGESHI*, ET. AL.

2026

Deep Research (DR) systems lacked a formal definition despite the emergence of several proprietary and open-source models. This work established a concrete definition of DR, introduced LiveDRBench as one of the first DR benchmarks along with a recursive evaluation framework. Highlighted performance gaps in DR systems from OpenAI, Perplexity, and Gemini.

interwhen: A Generalizable Framework for Verifiable Reasoning with Test-time Monitors

NeurIPS

V. BHAT*, P. CHANDA*, V. EKBOTE*, A KHANDELWAL, M. SWAROOP, ET. AL.

2026

Interwhen is a test-time verification framework for reasoning models to asynchronously verify intermediate reasoning. We improve reasoning soundness across code generation, logical reasoning, and agentic settings, without the high token overhead of standard test-time scaling methods.

Benchmarked vision-language models on inferring the message in rhetorical and atypical imagery used in advertisements, through three novel tasks. Found that VLMs struggle with reasoning about atypicality, but simple methods enable atypicality-aware verbalization and improve rhetorical image understanding.

Research Explorations

- Measured monthly Jensen–Shannon divergence in English Wikipedia visit and edit distributions, revealing a steady rise in entropy and growing randomness in collective information.
- Simulated event propagation on a graph-based SIR model to study how new events increase uncertainty and drive the observed rise in entropy.

- Designed a semi-supervised segmentation and graph-based VQA system for the [FloodNet challenge](#), combining CutMix and [Cross Pseudo Supervision](#).
- The VQA module uses geodesic dilation and morphological operations on segmentation maps, followed by reasoning over 4-adjacency graphs to count connected components.

Academic Service

2026	Reviewer , NeurIPS	
2025	Reviewer , ICLR Proceedings	
Fall 2023	Teaching Assistant , Operating Systems, ~250 students	BITS Pilani
Spring 2023	Instructor , Introduction to ML and DL, ~40 students	BITS Pilani
Fall 2022	Teaching Assistant , Object-Oriented Programming, ~250 students	BITS Pilani